TORCH EXPERIMENT

Let's Explore & Experiment!

WHAT IS HARDWARE?

Hardware is anything you can touch and hold! Tools, machines, and gadgets are all hardware. If you can pick it up and use it, it is hardware!

Hardware includes tools like screwdrivers, batteries, and torches!

We use hardware every day to make our lives easier and brighter!

WHAT IS A TORCH?

A torch is a small handheld light that we can carry anywhere. It runs on batteries and has a bulb or LED that glows brightly. Torches are very useful tools — they help us see when it is dark or when the power goes out!

A torch helps us see in the dark!

PARTS OF A TORCH

A torch has several important parts that work together to make light. Let's learn about each part and what it does!

Battery
Stores energy and powers the torch.

Bulb or LED
Lights up when electricity flows through it.

Switch
Turns the torch on and off.

Torch Body
Holds all the parts together safely.

HOW DOES A TORCH WORK?

Let's Find Out!

When you press the switch, electricity from the battery travels through the wires and lights up the bulb! Energy flows in a loop called a circuit.

No electricity = no light. Close the circuit = bright light!

MATERIALS NEEDED!

Gather these items before we start our torch experiment. Ask a grown-up to help you collect everything safely!

Torch body, battery, bulb or LED, and wires.

Cardboard or plastic tube for the torch body.

A switch, tape, and scissors. (Ask an adult to help with scissors!)

Keep your workspace clean and tidy before you begin!

STEP 1 - PREPARE THE TORCH BODY

Take the torch body — this can be a cardboard tube or a plastic tube.
Use a pencil or a sharp object to carefully make a small hole at one end of the tube.
This is where the bulb will go!
Make sure the hole is just the right size — not too big, not too small.

⚠️ Safety Tip: Always ask an adult for help when making holes or using sharp tools!

STEP 2 - CONNECT THE BATTERY AND BULB

Use wires to connect the battery to the bulb. Twist the wire ends tightly to the battery terminals and to the bulb holder. Make sure the connections are firm so electricity can flow properly.

Always check your connections before testing! Ask an adult to help.

STEP 3 - ADD THE SWITCH

Connect the switch so you can turn the torch on and off.
Attach the switch between the battery and the bulb using wires. Make sure the switch clicks when you press it!
Ask an adult to help you fix the switch firmly in place.

A switch controls the flow of electricity — on means light, off means dark!

STEP 4 - TEST YOUR TORCH!

Press the switch and see the bulb light up! Hold your torch and flip the switch. If it works, great! If not, check your wires and connections and try again!

Press the switch and watch the bulb glow bright!

If it doesn't work, check your wires and try again!

WHAT HAPPENS INSIDE THE TORCH?

Electricity Flow

Electricity moves from the battery through the wires to the bulb. The bulb lights up! Energy travels in a loop called a circuit.

Battery → Wires → Bulb = Light!

SAFETY TIPS!

Stay safe during the experiment! Follow these important rules to have fun and stay protected while building your torch.

- Always ask an adult for help with wires and scissors.
- Don't use old or damaged batteries — they can leak!
- Keep your workspace clean and tidy at all times.

FUN FACTS ABOUT TORCHES!

Did You Know?

Torches have a fascinating history and are super useful in our daily lives. Let's explore some amazing facts!

🕯️ The first torches used candles for light!

💡 Modern torches use LEDs that last a very long time.

🏕️ Torches help us during camping and power cuts!

LET'S RECAP!

Let's see what you remember about torches! Answer these fun questions and show what you know!

What are the main parts of a torch?

What makes the bulb light up? Why do we need a switch?

What should you do before starting the experiment?

GREAT JOB!

You did a wonderful job learning about torches! You now know the parts, how electricity works, and even built your own torch. That is amazing! Keep experimenting and never stop learning!

YOU ARE A SUPERSTAR SCIENTIST!

Keep exploring and stay curious!

THANK YOU & NEXT STEPS

Well Done, Scientists!

Thank you for learning about torches and hardware today! You did a wonderful job. Keep exploring and stay curious. Ask a parent or teacher to help you try more simple circuits at home!

Try making other simple circuits at home with help!